

SURVEY

A Comprehensive Review on Cybersecurity of Digital Twins Issues, Challenges, and Future Research Directions

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ABSTRACT This systematic literature review examines existing literature to determine how Digital Twin (DT) technology can enhance security, with an emphasis on emerging issues, approaches, and proposals in cybersecurity. The review categorizes selected studies based on the techniques used to protect DTs: Machine Learning (ML) and non-machine learning techniques. DTs are classified into five types: component level, process level, asset level, system level, and network of systems level, each with unique applications and corresponding cybersecurity threats. The main conclusions highlight medium to very high perceived risks depending on the DT type and industry implementation. Results identify the most common types of attacks on DTs, various security enhancement techniques, and the sectors that utilize DTs. Future research should focus on developing more effective security paradigms, technologies, and integration schemes to reduce risks and maximize DT technology's potential. This review presents the current status of DT-related cybersecurity and explores ways to enhance security against emerging cyber risks.

INDEX TERMS Digital twin, cybersecurity, virtual replicas, real-time monitoring, data protection.

I. INTRODUCTION

The DTs can be considered as revolutionary technology that connects the physical and virtual realities. A DT refers to a virtual replica of a physical object/structure, business, process or system created for real-time analysis and forecasting. This innovation has it used in many fields such as production, health, city planning, and energy among others as it has improved on observing, analyzing, and controlling various systems [1]. The fact that it is possible to run the working conditions' tests in virtual environment that helps to enhance the productivity and explore new opportunities [2]. Nonetheless, it is important to understand that while the adoption of DTs has the potential to positively transform industries, it also comes with significant cybersecurity risks.

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First, the control of the physical systems relies on the integration with their digital counterparts, meaning that there is the constant transfer of relevant data see Figure 1. This constant stream of data creates several threats which can be understood as weaknesses exploitable by hackers. It is thus crucial to protect DTs from threats in terms of the models and simulations, as well as the assets they represent. If proper security is not implemented in DTs, the smart replicas may experience cyber-threats which would result to disruptions and loss of data [3].

Some of the reviews focus on the general uses and developments of DTs. For instance, Enders and Hoßbach [4] provided a discussion on several dimensions for applications of DT based on which, the authors underlined their applications in various businesses. Semeraro et al. [5] offered some aspects and recent opportunities of the DT paradigm, particularly key concepts. Likewise, Jones et al. [6] have defined DTs from both conceptual and technological viewpoints and,

Singh et al. [7] have elaborated on the potential uses of DTs in numerous domains including healthcare, manufacturing, and smart city. In their article, Liu et al. [8] described the interfacing of DTs with industrial processes and the possibilities of their impact. These studies give a basic appreciation of DT technologies and their usage. But, it of course provides significant information on DT advancements and does not include a detailed study on specific cybersecurity threats of DT systems and their impact on the different sectors. More recently, due to the incorporation of DTs with the IoT, AI, and cloud technologies the possibilities of cyber threats were high for most industries including healthcare, manufacturing, energy as well as construction industries. Some of these risks are risks in the transmission of data, data storage threats, risks associated with authentication, and risks in emerging threats, for instance, ransomware as well as advanced persistent threats.

Challenges in the cybersecurity of DTs are diverse and specific to several fields. These include the weaknesses dealing with data transfer and storage, system identification, and interface with other structures built on the information domain. Integration of the DTs raises the level of vulnerability to cyber threats that may result in unauthorized access to, data leakage, or lack of proper operations of, the DTs. These are the challenges that need to be considered in order to implement an efficient security strategy designed for DT technology. Mitigation of these cybersecurity problems require application of various measures which are aimed at shielding DTs from vulnerability threats. These strategies include, the use of encrypted methods, tight and secure protective measures and constant supervising. Furthermore, approaches have to be flexible with regards to the context of various application fields and settings of DTs. This flexibility will guarantee that the security mechanisms are functional in what seems to be a dynamic technological environment [9]. The prime foundation for protecting DTs is to employ the best practices based on the established norms and existing findings in the literature. Such practices encompass security audits periodically, effective risk management, and integration of advanced technologies into an organization's cybersecurity measures.

This study aims to extend existing research by focusing specifically on the cybersecurity challenges faced by DTs in various application domains, identifying strategies tailored to these risks, and proposing actionable solutions.

From the best practices of case studies and scholars' advice, it is possible to get the best strategies to protect DTs and associated applications. Thus, in light of these best practices, risks may be avoided, and the effectiveness and security of DT technologies improved [10].

The organization of this paper is as follows: Section II introduces the methodology of the systematic literature review, detailing the criteria and methods used for selecting the papers included in this study, along with the survey questions that the SLR aims to answer. Section III highlights the research objectives. Section IV presents the systematic

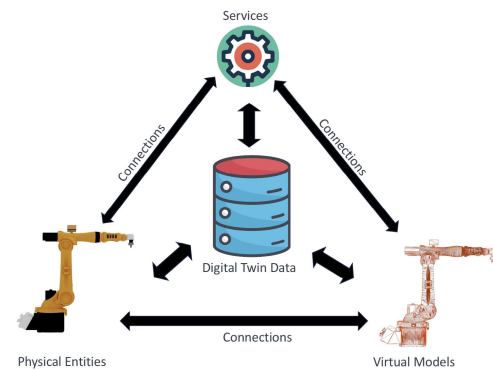


FIGURE 1. DT communication.

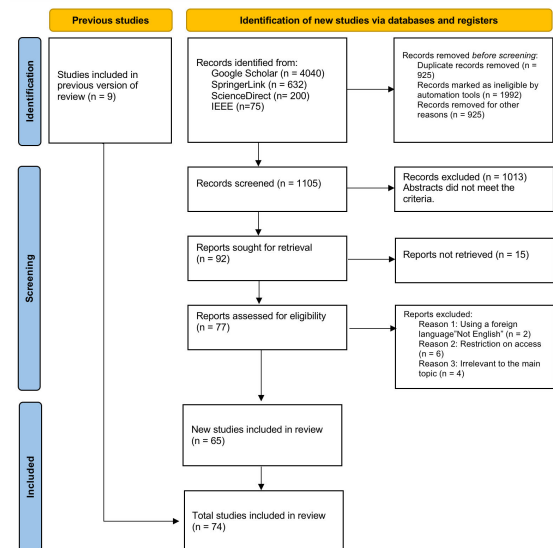


FIGURE 2. Selection of papers for literature review using PRISMA.

literature review, where various research papers are examined to identify challenges, corresponding strategies for protecting DTs, and section V shows a discussion of the results.

Section VI offers a comparison of our findings with those of other studies. Finally, Section VII highlights unresolved challenges and open issues that require further investigation, while the conclusion provides a comprehensive summary and discusses the implications of the research findings.

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II. SURVEY METHODOLOGY

The methodology for this systematic literature review on the cybersecurity of DTs involves several key steps to ensure a comprehensive and rigorous analysis, conducted following the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) approach [11]. Visual representation of the PRISMA methodology is illustrated in Figure 2.

A. PLANNING PHASE

The search of the literature performed using the given questions as keywords, keeping the focus on the cybersecurity of DTs and pointing out the problems, approaches, and recommendations associated with the topic. Since this is a review, it will focus on the literature published in the past 10 years to compare the innovation to the most current findings. These are articles in scientific and refereed journals, papers presented in learned conferences, industry documents and papers, and technical papers.

- RQ1: What are the specific cybersecurity challenges associated with DTs, including vulnerabilities related to data transmission, system authentication, and integration with other digital infrastructures?
- RQ2: How effective are current cybersecurity strategies in safeguarding DTs, particularly in terms of encryption techniques, authentication protocols, and monitoring systems?
- RQ3: What are the best practices for securing DTs, based on industry standards, empirical research, and successful case studies?

B. SELECTION PHASE

The choice of databases will be more comprehensive, so the mentioned resources will include IEEE Xplore, SpringerLink, ScienceDirect, and Google Scholar. These databases offer sound academic information and quality works with a comprehensive touch on the academic sides. Besides academic sources, the databases and journals related to the specific industry of study like Gartner, Forrester, and TechCrunch will be sought for existing latest practical information. Other types of sources also would be discovered in the catalogs of academic libraries, which could not be searched on the Internet. The main keywords will consist of the exact name of the concept “digital twins”, “cybersecurity”, “challenges”, “strategies” and “best practices”. Other keywords that would extend the scope of the search will comprise of “data security”, “risk management”, “encryption”, “authentication”, “continuous monitoring” and “vulnerabilities”. The Boolean operators (AND, OR, NOT) will help in narrowing down the search results and relevance of the information to be retrieved.

C. INCLUSION AND EXCLUSION CRITERIA

Based on the population criteria selected, the literature review will select studies that are targeted towards cybersecurity related to DTs. Sources include scientific papers such as journals, conferences, and reports including technical papers based on industrial surveys. Research papers in the English language only will be used as the focus to filter out ambiguities in translation. Sources with high levels of methodological stringency and reliable authors will be mostly used. On the other hand, any research not concerning the DTs or cybersecurity will not be considered. Other forms of articles such as magazine articles, editorials and other

articles that are not research based will also not be used. Also, only English language journals will be included, theoretical articles and those published within ten years from the time of writing will be also selected, while others will be excluded, however, any work that is considered as classic in that profession will be included regardless of when it was written. Publications that were not written in English will be excluded for the sake of having a standard language for comparison. We use (Mendely) which is a powerful tool that include but not limited to organizing large number of papers, it helped in in excluding duplicates and overlapping papers.

D. DATA EXTRACTION AND ANALYSIS

Data extraction shall identify relevant information such as the study’s aims and the objective as well as an outline of the study’s design and approach towards the topic of DTs cybersecurity.

To further process the extracted data, the technique applicable will be thematic analysis where the data will be grouped into themes such as challenges, strategies, as well as the best practices. Hence, a patterned and experimented method of data analysis will be followed to make understandings about the cybersecurity for DTs recurring and more explaining comprehensively. Descriptive analysis will be employed with the aim of comparing studies based on their characteristics within the domain and contrasting them in other aspects. Within the synthesis process, findings shall be integrated to unearth related trends and gaps in the literature. We use the (VOSviewer) tool to help in analyzing the data and categorizing the research findings into different themes or clusters, which allows us to identify key trends, relationships, and gaps in the literature, See Figure 3.

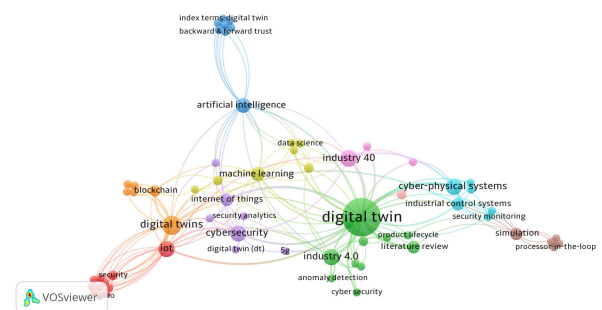


FIGURE 3. VOSviewer visualizing network for DTs’ selected papers.

III. PROBLEM STATEMENT

Indeed, as the concept of DT technology is continuously implemented and incorporated in different business fields, the security threats for the systems are on the rise as well. There are several vulnerabilities related to security that are connected with DTs, which collect and transfer large amounts of sensitive data between real and cyber worlds. They include data vulnerability, unauthorized access and interruption to operations and organization’s IT facilities. Despite the huge

potential DTs have in increasing the capability in monitoring, simulation, and optimization, the implementation of DTs has not incorporated adequate cybersecurity that threaten the effectiveness of systems and can potentially compromise critical infrastructure. Therefore, it is imperative to solve these cybersecurity concerns so as to investment in effective solutions about DTs technologies.

A. RESEARCH OBJECTIVES

- 1) To study and identify the specific cyber threats association with DTs including the issues concerning the data exchange, system identification, and integration with other frameworks.
- 2) To analyze cybersecurity measures applied to protect DTs and evaluate the outcomes, and key activities on encryption and decryption, authentication, and monitoring.
- 3) To determine the main approaches or principles of safeguarding DTs and develop a reliable cybersecurity approach based on industry benchmarks, scientific findings, and business success stories.
- 4) To provide a set of practical suggestions for organization to use in improving their security status to accommodate DTs which will use virtual components as well as their physical counterparts, guarding against possible cyber threats.

IV. SYSTEMATIC LITERATURE REVIEW

A. CHALLENGES IN DTs CYBERSECURITY

DTs stands for a complex blending of the physical and cybernetic infrastructure, which involves the creation of copies of the real-life resources or the actual production processes. Although this innovation has many advantages, it has at the same time introduced in several cybersecurity issues that have to be considered in order to protect such systems [12]. However, one of the major issues in DTs cybersecurity is associated with the data transmission issues. DTs operate based on endless streams of information flow between the physical and their corresponding digital counterparts. This non-stop influx of information can also prove to be vulnerable to interception, modification, theft or even denial of service (DoS) attacks. For instance, if someone gets access to the information exchanged between, say, a car and its virtual counterpart, they might alter the data feed, for example, to sabotage the car or steal the information. To ensure such data flow is temporarily or permanently sealed with appropriate means, it is crucial to use reliable encryption means; however, this also increases the system's complexity [13]. Another considerable problem is system authentication. DTs are often created with the participation of users and connected devices and other elements of digital environment. It is paramount to guard the DT so that only the intended parties can read or write into it. However, their authentication may not suffice based on the conventional methods because such systems are dynamic,

and highly integrated. To deal with these challenges the use of second factor authentication and biometric systems is required but their adoption is costly and requires more resources. Connection with other digital structures also brings new possibilities of cyber threats. It is customary to use DTs in conjunction with extended digital environments like the cloud, the IoT, and in-house systems. Such integration leads to creation of multiple access points which can be used maliciously by the hackers. Every connection or interface creates risk, and it becomes difficult to reduce the risks that affect the entire system. It is necessary to note that the issue of cybersecurity is critical for DT architecture since all the components encompassing the infrastructure must be protected together with the data being exchanged between them [14]. Another challenge from the Emerging threats and trends is observed in the context of DTs cybersecurity. Since now, with Artificial Intelligent (AI), and emerging technologies, there are new patterns of cyber threats that may exist and that may be developed for DT systems uniquely. For instance, inherent threats such as ransomware and Advanced Persistent Threats (APTs) change over time and becoming more complex, it implied that the DT systems must be equally enhanced and adapted to meet the emerging threats. It is very vital to consider the above listed threats not only in computer technology but also in the overall security strategy. Solving these problems calls for a complex approach based on the use of consistently enhanced protection measures, operative detection, and understanding of specific threats relating to DTs. In addressing these issues explicitly, one can foster security and safeguard both physical and cyber DT systems of an organization [15].

To illustrate the key findings on how cybersecurity threats and risks are realized across a broad set of industries employing DT technologies, Table 1 provides a summary of security concerns and their impact when adopted in different sectors.

B. APPROACHES TO PROTECT DTs

DTs' protection implies a multi-layered strategy due to the nature of the objects and their specificity that arose in connection with the development of highly interconnected systems. Here are key strategies for enhancing the security of DTs:

- 1) Encryption Techniques: Security of the data communicated between physical objects and their electronic analogs starts with encryption. Encryption involves transforming data in a way that the message or information which is highly confidential cannot be used by other people since it will be in another form which is not understandable. Encryption of data both in storage and in transmission should always be launched. Therefore, for DTs, it entails employing encrypted standards AES and safe approaches in conveying data such as TLS/SSL [22].
- 2) Authentication Protocols: Secure user authentication measures are critical to guaranteeing that only the

TABLE 1. Comparison of security issues in digital twins across industries.

Sector	Security Issues	Description/Examples	References
Healthcare	Patient data privacy, compliance with Health Insurance Portability and Accountability Act (HIPAA)	Unauthorized access to sensitive patient data in DT healthcare systems.	[9]
	Real-time monitoring vulnerabilities	Exploitation of IoT devices in patient monitoring.	[16]
Industrial Applications	Ransomware targeting operational technology	Attacks halting production in manufacturing systems using DTs.	[3]
	Cyber-physical system (CPS) risks	Malicious control of machinery via DT interfaces.	[17]
Energy	Smart grid security	Hacking into DTs managing energy grids, causing outages.	[18]
	Intrusion into renewable energy systems	Unauthorized access to wind farm DTs.	[19]
Construction	Integration with BIM	Security risks from insufficient DT-BIM authentication.	[20]
	Data integrity	Manipulation of resource data within construction DTs.	[21]

right user's or systems get access to and interact with DTs. Conventional ID and authentication techniques in the form of simple usernames and passwords cannot hold the ground in DT business because of the criticality and sophistication involved in the twin surroundings. Additional protocols such as the Multi-Factor Authentication (MFA), which the user has to authenticate with a second layer such as passwords, biometric data, and/or security tokens just to be granted access, increases the layers of security. Furthermore, there is the use of proper authentication control for the devices and systems within DTs to meet and deter the intrusion and breaches [23].

- 3) Continuous Monitoring Systems: Real-time response or detection is essential for security threats especially in computers and cyber security. For DTs, this refers to use of sophisticated tools with elaborate capabilities of monitoring the system activities to detect anomalies and inform the administrator of any security threats. Intrusion Detection System (IDS), Intrusion Prevention System (IPS) and Security Information and Event Management (SIEM) solutions should be included in the solutions that a monitoring program should embrace. These tools assist in preventing the threats so that they do not result in substantial loss [24].
- 4) Adaptive and Flexible Security Strategies: Due to the ubiquity of DT and systems it occupies various application domains and environmental contexts, the security strategies must be sufficiently malleable. It means that the security measures should be adjusted to the needs and challenges related to a particular DT application. For instance, there is a difference in the level of security that a DT for a smart city use case will require as compared to a manufacturing use case. Thus, the risk-based approach that consumes less time and resources but focus on identifying and applying security measures according to the criticality and sensitivity of the DT [24].
- 5) Regular Security Assessments: Security audits should be carried out periodically to determine the weaknesses and check the efficiency of implemented security measures. Some of the assessments that should be conducted include the vulnerability assessments, penetration testing, and security audit test. This approach means that by constantly scrutinizing and modifying the current security paradigms, threats and vulnerabilities can be dealt with and changes in the technological environment minimized. It also aids in the compliance with industry standards and regulatory requirements of the business through security assessments done from time to time [25].
- 6) Risk Management Frameworks: Discussed measures of carrying out risk management for DTs aid in methodically recognizing, evaluating, and managing risks. Actually, there are various standards which help in managing cybersecurity risks for instance the NIST Cybersecurity Framework or the ISO/IEC 27001. These frameworks provide organizations with general guidelines for the formulation of policies, procedures, and controls that will safeguard DTs as well as assure that risk management is a consideration across a firm's operations [26].
- 7) Adoption of Advanced Technologies: The following ways can be used to improve the security of a DT through advanced technologies: For example, AI and ML can be applied in enhanced threat detection and handling of threat since the two technologies can sort through humongous data sets and come up with patterns that indicate threats. Since data stored in each blockchain should be impeccable and verified by other participants in the distributed network, blockchain can offer extra layers of safety within the identification of the transferred data's authenticity by way of DTs [27].
- 8) Industry Standards and Guidelines: Staying updated with the policies enhances the practices in organizations to conform to standard professional guidelines and relevant policies. References like the Industrial

Internet Consortium's (IIC) Security Framework or the International Society of Automation's (ISA) 62443 series offer good outlines for securing DTs. Adherence to these standards aids in the creation of proper security systems and the recognition of stakeholders' confidence, when using the mentioned strategies, the organization can establish a strong security perimeter around the DT systems and guarantee their dependability and credibility [28].

C. DIGITAL TWINS SYSTEM LEVEL SECURITY

Table 2 categorizes digital twin types alongside with their cybersecurity risk levels into five distinct DT types: The component, process, asset, system, and network type. As we can see each tier is associated with the particular applications and security threats showing the distinctions in the risk levels. DTs' risks are assessed as moderate to high severe level depending on the type of DTs' exposure to threats. Such vulnerabilities, in turn, can be exhibited by inaccurate data generation, operational failures, and untoward incidents. As for component type, the risks range from medium to high because main elements, such as sensors and actuators, which act as a bridge between physical and digital systems, are likely to be affected. For example, malicious sensors in a manufacturing environment may send erroneous information which affects the entire system. Mullet et al. [17] thus state that organizations need to incorporate stability features such as secure communication and methods used for encrypting data to counter such failures. At the process type, DTs enhance the workflow and monitor processes that take place in real time. These applications are considered medium and high risk since data can be altered by unauthorized users or the system stops working. In worst-case scenarios, such occurrences could lead to operational issues or emergencies in business activities for example construction, manufacturing industries. The drawbacks include problems such as low reliability and improper data handling, which Huo et al. [29] solve through blockchain improvements in integrity and transparency to protect key processes. At the asset type, DTs operate sophisticated physical plant equipment like an excavator, crane, or an industrial robot among others. Such systems are in greatest danger because unauthorized access to these DTs may result in physical failures that threaten safety and result in financial losses. Hasan et al. [30] have argued that to secure these high-value assets there must always be effective access controls and authentication put in place. The ones that operate at the DT system type and handle managing vital systems, like electric power, or industrial systems, have extremely high risk rates. These systems are especially vulnerable to cyber threats due to their complexity; a breach would likely cause mass operational failure and maybe endanger lives. For instance, Cioara et al. [18] have proposed a set of security mechanisms, including constant surveillance and multi-factor identification as key in mitigating the threats against the

DT system type. Finally, at the network type, the DTs are most complex and interrelated and hence threatening to cybersecurity. These DTs connect several systems from different domains, and even a minor vulnerability means severe outcomes, including mass service disruptions, significant financial losses, and devastating impacts on an organization's reputation. Atalay and Angin and Khan et al. [19], [31] discussed that to address these risks, there is a need to develop and implement modern intrusion detection systems as well as secure communication protocols. For example, DTs on the network type in the case of the smart grid, must be based on highly secure frameworks that prevent the disruption of the grid and chain reactions. By recognizing those different levels of risk and the challenges associated with each, one can make proper cybersecurity plans. The following are essential in the protection of DT systems: Such solutions include the use of Blockchain, AI-based threat detection, as well as the use of adaptive security frameworks. Such targeted strategies allow for maintaining the reliability and confidentiality of the use of DT technology in different fields.

D. STUDIES PERIMETER AROUND THE DT

Securing communication between DTs and Industrial Internet of Things (IIoT) deployments is crucial, especially in critical infrastructure, as DTs rely on real-time data from IIoT devices to function effectively. Ensuring the integrity and confidentiality of this data is vital to prevent cyber threats and maintain operational safety, see Table 3. Research highlights various security measures: Mullet et al. [17] emphasized the need for comprehensive cybersecurity guidelines to address gaps in current practices and safeguard communication channels. Huo et al. [29] discussed the role of blockchain technology in enhancing data integrity and transparency, which is essential for securing data exchanges between DTs and IIoT devices. Homaei et al. [32] explored the application of artificial intelligence (AI) to reinforce cybersecurity, using ML to detect and respond to threats effectively. Additionally, Saracco and Henz [33] and Stjepandić et al. [34] contributed to understanding DTs and their applications, underscoring the need for secure communication to optimize industrial processes. Overall, integrating robust cybersecurity practices, blockchain technology, and AI is essential to protect DT systems and ensure the reliability of data exchanged between virtual and physical entities.

In accordance with Gourisetti et al. [35] the digital twin environments required the establishment of comprehensive verification and validation processes to effectively mitigate potential threats, including data corruption, and ensure their integrity. However, it was imperative to carefully assess the trade-offs, such as latency and feasibility issues, especially when dealing with specific systems like those in the energy sector. Furthermore, the study revealed that approximately 27% of the examined use cases in DT-based analyses aligned to enhance cybersecurity through detection and forecasting. Security issues in these environments primarily

TABLE 2. Classification of papers into five digital twins types and associated cybersecurity risk levels.

DT Types	Focus	Risk Level	Cybersecurity Risks
Component	Focuses on smart manufacturing components such as sensors and actuators for monitoring and control in real-time systems.	Medium to High (MHR)	Failures in components like sensors or actuators could lead to system-wide disruptions.
Process	Focuses on site monitoring to automate and optimize construction processes.	Medium to High (MHR)	Disruptions could lead to delays and safety incidents.
Asset	Focuses on complex construction equipment and machinery interaction (e.g., dozers, excavators).	High Risk (HR)	Unauthorized access or failures could lead to machinery malfunctions and safety risks.
System	Focuses on managing large and interconnected systems like smart energy grids.	High Risk (HR)	Failures or breaches could lead to large-scale operational disruptions.
Network	Focuses on managing and optimizing distributed wireless communication systems.	Very High Risk (VHR)	Breaches could lead to widespread communication failures and network disruptions.

revolved around a cybersecurity gap between the digital replica and the physical system, which could potentially provide adversaries entry to the digital twin. To handle these concerns, security mitigations included implementing a robust and well-structured approach to secure software development, automated peer code reviews, the adoption of secure coding methodologies, enhancing security through proactive detection and predictive analysis, conducting vulnerability assessments, and implementing measures to tightly control access to software and data, limiting them to specific authorized devices. In addition, the paper includes many detailed diagrams that are complex and difficult to understand. It is suggested to redraw these diagrams to make them simple to readers, without changing their content.

The study by Stergiou et al. [16] addressed critical concerns in a healthcare digital twin scenario. Leveraging the IoT and advanced data management techniques, their research introduced a novel approach to safeguarding sensitive healthcare data, contributing to enhancing patient care and data security in the healthcare digitization landscape. In 2023, their study specifically focused on healthcare and identified several security issues, including data breaches, vulnerability to exploitation, potential misuse of AI, privacy implications, and the need to address bias. To mitigate these concerns, the study proposed security measures such as data encryption, access control, ensuring data integrity, regular updates, compliance with regulations, monitoring and auditing practices, providing education and training, and establishing an effective incident response system. The suggestion for the paper is to include detailed quantitative security measures to support its theoretical proposal.

Atalay and Angin [19] introduced an innovative perspective on addressing security concerns in the smart grid domain. They highlighted the vulnerabilities and threats faced by smart grids and put forward a groundbreaking idea of using digital twins to facilitate security testing without causing service disruptions. This digital twin approach presented a promising solution to the security challenges associated with critical infrastructure in the energy sector. It laid the foundation for standardized models that supported constant and comprehensive penetration testing of smart grids. The study, conducted in 2020 within the energy sector, primarily focused on the vulnerability of smart grids to cyberattacks.

Smart grid attacks can cause extreme outcomes, such as damage to critical infrastructure. To mitigate these security issues, the paper emphasized the adoption of digital twins, enabling security testing in a non-disruptive manner. This approach allowed for thorough and continuous security testing to identify vulnerabilities and threats in a controlled environment, ultimately enhancing the security of smart grids. The suggestion for this paper is to include relevant review papers on the subject to strengthen its claims about the use of digital twins in smart grids.

Suhail et al. [36] introduced sEcuriNg dIgital twins through GaMification Approach (ENIGMA), a security measure for cyber-physical systems with digital twins. The paper highlighted potential security risks, including unauthorized access, data breaches, and cyberattacks, while also addressing the lack of transparency in digital twin systems. ENIGMA, developed for the IT industry, offers an explainable security solution. It utilized gamification techniques to raise cybersecurity awareness and provides a transparent method for identifying and mitigating security vulnerabilities in digital twin systems. The paper also used Deep Neural Networks (DNN) adopting two datasets Survival Analysis Dataset (SAD) and Oak Ridge National Laboratory (ORNL) to enhance training in digital twin system while using SHapley Additive exPlanations (SHAP) for feature selection. These findings and ENIGMA's implementation offer practical security solutions for various sectors employing digital twin technology. The suggestion for this paper is to include standard performance metrics, such as accuracy, precision, and F1-score, in addition to comparing their results with other existing studies.

Empl and Pernul [37] developed and implemented the Digital Twin-based Security Analytics (DT2SA) model, aimed at enhancing cybersecurity in the IoT within the IT industry. Their work emphasized how DTs could improve security analytics by enabling meaningful semantics and bi-directional communication with physical counterparts. The document also included a case study of implementing this model in a microservice architecture called TWIN-SIGHT. DT2SA tackled key issues such as data privacy and cyberattacks. For these challenges, the authors recommended employing encryption for data storage and transfer, limiting data access to essential personnel, utilizing advanced

cybersecurity measures, including firewalls and intrusion detection systems, and maintaining regular system software updates. Their innovative work demonstrated how DTs can play a pivotal role in improving cybersecurity knowledge sharing and alignment with digital twins, offering a practical model for the industry. The suggestion for this paper is to include more research papers discussing the security of digital twins in general.

Dong et al. [38] discussed designing and implementing a digital twin diesel generator system for research and educational purposes. The document also explained how to use MATLAB/Simulink or MP2Link to build and implement the digital twin system and how to use the system for real-time online monitoring and data-driven modeling. It emphasized the system's benefits for comprehensive remote experiments and for enhancing the accuracy of mathematical models. The suggestion for this paper is to include security considerations, such as secure authentication, encryption, and access control mechanisms.

Eckhart and Ekelhart [39] introduced a framework designed to implement digital twins in the context of ICS, referred to as CPS Twinning. This framework aimed to enhance the security of critical infrastructure systems. It accomplished this by making a simulated, virtual representation of the physical environment. This virtualization allowed for comprehensive testing and monitoring of the system, all without posing any risk to the actual physical infrastructure.

The document concentrated into the framework's architecture, outlined the techniques employed for generating DTs from system specifications, and offered a practical proof of concept for the framework's implementation. The research focused on addressing issues associated with the inherent vulnerability and insufficient security measures in CPSs and ICSs. These vulnerabilities can potentially expose critical infrastructure to cyber threats and attacks. To avoid these security challenges, the authors suggested system detection and testing. The suggestion for this paper is to support more industrial communication protocols because it only uses Modbus TCP/IP.

Bécue et al. [40] presented a novel idea of a digital twin for optimizing and enhancing the resilience of future factories. This approach addressed two critical limitations of existing technology: the need for techniques to model human-machine interactions and the closed-system nature of digital twins, which slows holistic process optimization. The paper also explored the potential of DTs in achieving cyber-resilience objectives and proposed convergence with security simulation tools known as Cyber-Ranges. The study showcased a comprehensive factory of the future modeling and simulation framework, leveraging tools like Gazebo, Airbus Cyberange, and Elasticsearch to merge physical and cyber security data. Notably, the paper emphasized enhancing the optimization and resilience of smart manufacturing infrastructure, which is underscored by the additional benefits of incorporating

human behavior modeling and co-simulation. The suggestion for this paper is to clearly compare the virtual framework with the real system, showing their differences in performance and behavior for better understanding.

Cui et al. [1] developed a control system for a Multi-Agent System (MAS) using an AI assistant. The system is designed to withstand composite attacks and ensure secure agent communication. The research outlines strategies for defense against Denial of Service (DoS) attacks and unbounded actuation attacks, using a layered approach with a Twin Layer (TL) and a Cyber-Physical Layer (CPL). The author also discussed the system's use of distributed observers and adaptive controllers. The AI assistant is programmed to follow specific guidelines and protocols to ensure its identity and the identity of OpenAI are concealed during interactions. The suggestion for this paper is to include a performance comparison with other existing methods to clearly demonstrate improvements and effectiveness.

In their discussion on the use of DTs in the Metaverse, Far and Rad [41] highlighted the difficulties related to user interface, security, and privacy. It began by providing an overview of DT and the Metaverse, along with its uses and general issues. Next, it suggested a three-layer architecture that would employ an interface to link the Metaverse with the physical world. It examined the privacy and security issues surrounding the use of DTs in the Metaverse. Additionally, the study offered potential fixes for the issues raised and made recommendations for more research in this area. The suggestion for this paper is to include detailed technical implementation guidelines and practical validation results to demonstrate the real world feasibility and effectiveness of the proposed architecture in Metaverse environments.

Sipola et al. [42] proposed of creating an automated replica of the food supply chain. A digital replica of the food supply chain designed for cyber range-based cyber security drills is described in the study. The design science process is applied in the creation of the digital twin, which precisely replicates the actual scenario. The system's primary drawbacks are that it is only implemented on one particular platform, and additional feedback from several cyber exercises is required. Cyber ranges can instruct enterprises involved in the food supply chain by building a digital twin. The suggestion for this paper is to reduce platform dependency by making the system adaptable to other platforms beyond RGCE and to include benchmark comparisons with other existing digital twin implementations to better evaluate its performance.

Holmes et al. [10] discussed the growing significance of DTs in the context of Industry 4.0 and the associated cybersecurity challenges. The paper outlined various security issues faced when implementing DTs and offers corresponding solutions. Notable security issues include financial constraints, data integrity, IP leakage risks, safety concerns, and confidentiality preservation. Solutions include improved patch management, advanced system testing, risk management enhancement, active cyber defense with

Cyber DTs, advanced training through digital twin cyber-range environments, privacy-preserving technologies, and potential blockchain adoption for secure transactions. The suggestion for this paper is to include real-world case studies to validate the proposed cybersecurity measures in practical digital twin environments.

Suhail et al. [43] discussed cyberattacks on DTs within CPS. They explored attacker strategies, the role of replication and simulation modes, and considerations for targeting the physical system or virtual twin. The paper emphasized using blockchain to secure digital twin data and introduce smart contracts for automation. It also proposed a gamification approach for security assessment, involving a controlled virtual training environment and intelligent incident response to improve digital twin security. The paper's objective was to improve and strengthen the security measures of DTs against various attacks and ensure the trustworthiness of data achieved through blockchain technology and gamification. The suggestion for this paper is to provide specific technical implementation details for the proposed security measures and to address ethical concerns, including privacy, data ownership, and access control, to ensure a more comprehensive approach to digital twin security.

Mylrea et al. [44] discussed the development of a BioSecure digital twin, a concept aimed at improving cybersecurity and efficiency in biopharmaceutical manufacturing. This digital twin is designed to enhance manufacturing resilience and preparedness against malware attacks, supply chain disruptions, and human errors. It enables real-time monitoring, integrity checks, and process control while preserving data lineage and provenance. The proposal also includes the creation of a BioSecure Observe Orient Decide Act (OODA) Loop Framework and a call for additional resources and expertise for relevant government departments. The document emphasizes the importance of advanced cybersecurity measures due to increasing cyber threats in the biopharmaceutical industry. The suggestion for this paper is to add training programs for stakeholders, such as operators and cybersecurity teams, to ensure they can interact with and manage the BioSecure digital twin.

Danilczyk et al. [45] discussed developing and applying the Automatic Network Guardian for Electrical (ANGEL) framework, which is founded on DT technology. The system is developed to handle security obstacles in the modern power grid, particularly cyber-physical security. It highlights the vulnerability of the power grid to cyberattacks and presents potential security issues like false data injection, denial-of-service, and topology attacks. The ANGEL framework is proposed as a solution to these challenges, offering real-time monitoring, analysis, and control of the power grid. The authors emphasize the importance of using this technology to safeguard critical infrastructure. The suggestion for this paper is to expand the literature review, compare ANGEL with existing frameworks, and validate it in real-world scenarios to better address cybersecurity threats and standards.

The paper by Christopher Carr et al. [46] explored the security and safety challenges in the context of digital twin systems for robots, focusing on Robot Operating System (ROS) driven systems. The authors highlighted the critical relationship between security and safety and discussed how attacks on digital twin systems can have severe consequences, including harm to humans and surrounding facilities. They provide case studies involving a TurtleBot 3 and a Universal Robot 10 to demonstrate vulnerabilities in ROS-based robotic digital twin systems, particularly in Person-in-the-Middle (PitM) attacks. The research emphasizes the need for robust security measures in ROS-driven robotic systems to ensure the safety of human operators in safety-critical scenarios. The suggestion for this paper is to include clear mitigation strategies for each identified attack type to provide practical solutions for preventing or responding to these vulnerabilities effectively.

Pokhrel et al. [2] discussed the concept of DTs and their application in various sectors, with a particular emphasis on enhancing the safety and security of cyber-physical systems. It outlines different tools and technologies associated with digital twins, including both commercial solutions like GE digital twin and open-source tools like Eclipse Ditto. The text explores how DTs can be utilized to detect security threats, predict anomalies, and improve the safety and reliability of equipment. It presents various use cases for digital twins, such as intrusion detection, predictive analytics, and virtual commissioning, with a focus on their role in enhancing security and safety across different industries. The paper should include measurable results like detection accuracy to make the findings clearer. It should also provide a clear roadmap for future research with specific challenges and goals.

Damjanovic-Behrendt and Behrendt [47] provided a comprehensive exploration of DT technology, specifically in the context of smart manufacturing. It outlined the essential components for creating a DT, namely data acquisition and analytics, models for data representation and computation, and services for connectivity and cybersecurity. The document demonstrated how open-source tools can be utilized to build an effective DT infrastructure. It also addresses challenges and outlines future research directions within this field. The paper should expand on ethical concerns, including data privacy, user consent, and governance, as these are critical for building trust and ensuring responsible implementation of digital twins in smart manufacturing.

Pavlov [48] discussed the security concerns and authentication techniques related to using DTs in IoT platforms. The paper provided an overview of the IoT background, the concerns related to authentication and security, existing authentication schemes, and recent proposals. It also provided insight into the DT technology, its history, specifications, current development, benefits, and challenges of its implementation. The paper included a simulation showing how DT technology can be deployed in the IoT platform and

how devices can be authenticated. The paper recommended various security solutions for IoT and DTs, including scalable authentication schemes for IoT architecture, accommodating device heterogeneity, combining software and hardware solutions, employing Physical Unclonable Function (PUF) technology, implementing secure authentication and key agreements, optimizing data packet encapsulation, and using biometrics with Elliptic Curve Cryptography (ECC). It also advocates distributed and hierarchical security approaches, a blend of encryption methods, PUF-based Authenticated Key Exchange, and a hybrid Diffie-Hellman authentication scheme. These solutions aimed to enhance the security and functionality of IoT and DTs. The paper should address real-world implementation barriers, including challenges such as legacy system integration, cost of deployment, and complexity in large-scale adoption, to ensure the proposed solutions are practical and applicable across diverse IoT environments.

Dietz et al. [49] discussed the idea of a digital twin, a digital model of a real-world object, and the challenges of securely sharing data bonded to the digital twin. The authors proposed distributed ledger technology to ensure data integrity and confidentiality in DT data sharing. The paper outlined the requirements for secure data sharing, described the system architecture, and presented a use case to demonstrate the functionality of the proposed framework. The paper should explain how data sharing is managed between parties, ensure off-chain data is secure, and explore flexible access controls. It should also address energy use and smart contract risks for a more reliable system.

Cathey et al. [50] introduced an innovative strategy for securing data in the IoT, employing multiple device shadows for a singular physical object. This method segregated data across distinct virtual entities, utilizing dynamically assigned tags to restrict access to specific data points exclusively for authorized users or applications. The presented solution is implemented at the edge, guaranteeing minimal latency and real-time protection measures with negligible overhead. It exhibits lightweight characteristics, as evidenced by the performance metrics obtained. It further outlined the implementation details and the corresponding performance metrics. In conclusion, the paper summarized the undertaken work and outlined opportunities for future advancements in this domain. These limitations highlight critical areas for improvement, integration barriers with legacy systems, synchronization delays, and governance gaps. Addressing these issues would enhance the system's robustness and practical applicability.

Jagannath et al. [51] presented a scalable intelligent DT framework for data-driven modeling and real-time simulation of extensive systems over 5G/6G-based IoT networks. The article looked into incorporating AI and ML engines for tasks such as event detection, predictive modeling, learning, and planning, encompassing various services and functionalities within a unified framework. Furthermore, the paper investigated the design and application of intelligent Zero-Trust

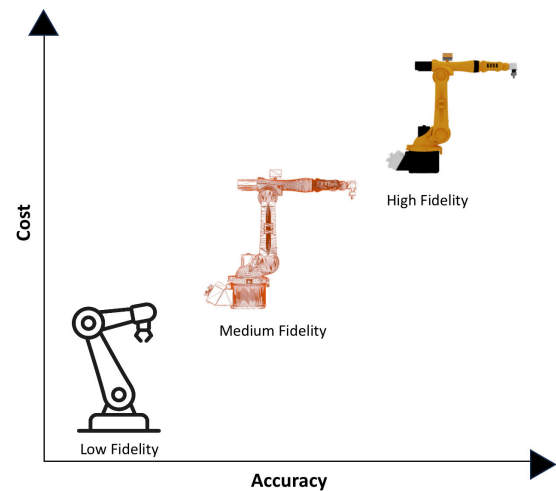


FIGURE 4. The fidelity of the DT system.

Architectures (ZTA) as a practical instance of the proposed DT, facilitating fast prototyping, testing, risk assessment, and the seamless integration of innovative security solutions into existing large-scale systems. The paper needs to explain how to reduce computational demands from advanced ML models and minimize latency caused by cloud-based synchronization to ensure smoother performance on resource limited IoT devices.

Assessing this collection of studies, the necessity of DTs as the virtual models of the associated physical systems and the diversification of their focus and uses appear obvious for the majority of the contributors. Thus, the definitions and conceptualizations of the DTs provided in the papers align with one another as focusing on the possibility of the model to reflect the physical object in the digital environment. Various fields of implementation start from increasing security in manufacturing [17] and the application of artificial intelligence in the sphere of cybersecurity [32] up to the analysis of blockchain potential in IIoT [29]. These technological components comprise blockchain, AI, and sensor networks for the IoT, each study examines various aspects of these technologies. Issues include lack of frameworks in cybersecurity, issues of AI adoption, and problems with existing DT solutions. Some directions for the further research include the improvement of technology, better adjustment of the integration with the blockchain, better AI techniques, and others. This overview also proves the assumption that the usage of DTs is capable of changing a number of industries for the better now being supported by real-life issues and future developments.

E. MACHINE LEARNING APPLICATIONS IN DIGITAL TWINS:

This subsection can focus on integrating ML techniques within the digital twins. It covers ML applications for predictive maintenance, anomaly detection, and other functionalities within digital twin platforms, see Table 4.

TABLE 3. Summary of literature review.

Ref.	Year	Sector	Tool	Key Findings/Security Issues
[17]	2021	Industry	Not specified	Analyzing existing cybersecurity guidelines for manufacturing in Industry 4.0. Identifying gaps and proposing best practices to protect against cyber threats.
[29]	2022	IT	Blockchain	Blockchain and DT integration challenges.
[33]	2022	IT	Not specified	Presented a collection of research papers on DTs, showcasing advancements in technology and diverse applications.
[52]	2019	CPS	Not specified	Explored the concept of DTs, detailing their technological foundation and potential to optimize physical systems through virtual representation.
[32]	2022	IT	AI	Investigated the application of AI techniques to enhance the security of DTs, including threat detection and response strategies.
[34]	2021	Manufacture	AI	Offered a foundational understanding of DTs, explaining their architecture, components, and potential benefits for manufacturing processes.
[35]	2023	Energy	Not specified	The existence of a security gap between the digital replica and the physical system. The potential for an adversary to acquire entry to the digital replica.
[16]	2023	Healthcare	Not specified	Data Breach. Vulnerability to Exploitation. Potential Misuse of AI. Privacy Implications. Addressing Bias.
[19]	2020	Energy	Not specified	The paper underscores the vulnerability of smart grids to cyber attacks, which can have severe consequences, including service disruptions and potential damage to critical infrastructure.
[36]	2023	IT	Azure, DNN, SHAP	The paper discusses potential security risks of digital twins, including unauthorized access, data breaches, and cyber-attacks. It also highlights the lack of transparency and explainability in digital twin systems, making it difficult to detect and mitigate security loopholes.
[37]	2023	IT	DT2SA	Data Privacy. Cyberattacks.
[38]	2023	Diesel generator system	Matlab Simulink and M2PLink	The author did not specify any security concerns.
[39]	2018	ICS	CPS Twinning	The inherent vulnerability and lack of adequate security measures for CPSs and ICSs, potentially exposing critical infrastructure to cyber threats and attacks.
[40]	2020	Aerospace System Manufacturing	Cognitive modeling	Exploiting Fail-Open mechanisms. Interoperability Concerns: Bridging the gap in DTs and cyber Ranges.
[1]	2023	IT	AI, Composite Attacks	Denial-of-services attacks. False-data injection attacks. Camouflage attacks. Actuation attacks.
[41]	2022	IT	Not specified	Integrity, Confidentiality, Availability.
[42]	2023	Food supply chain	Simulated model	Cybersecurity threats targeting food supply chains.
[10]	2021	IT	Not specified	Availability constraints due to financial limitations. Unauthorized modifications or destruction of data and operations. Risk of IP leakage and trade secret exposure. Potential threats to safety when using Cyber DTs for testing. Challenges in preserving confidentiality of sensitive information.
[43]	2021	IT	Not specified	Reconnaissance attacks. DTs Operation Modes can be exploited. Victimized Physical System or Twin. Drawbacks of Desirable Features: Higher fidelity in DTs can make them more vulnerable.
[44]	2021	IT	BioSecure OODA	Vulnerable operational technology due to digitization and automation, expanding the attack surface. Ineffectiveness of current cybersecurity defenses, especially for operational technology. Vulnerable proprietary protocols designed for intellectual property. Lack of basic cybersecurity defenses for critical assets. Extended detection time for cyber breaches. Resource-intensive tuning of new security tools. Inadequate regulatory guidance and compliance processes in addressing digital threats. Insecure data collection and auditing in interconnected systems.
[45]	2019	Energy	ANGEL framework	Vulnerability of the power grid to cyberattacks. False data injection, denial-of-service attacks, and topology attacks.
[46]	2022	IT	ROS	PitM attacks.
[2]	2020	IT	GE DT, Windchill, Eclipse Ditto, CPS Twinning	Security threats, intrusions, and anomalies.
[47]	2019	IT	Micro-services	Leveraging micro-services has the potential to introduce security weaknesses that may compromise the reliability of services.
[48]	2022	IT	Azure	Decentralized Mutual Authentication and Vulnerabilities. Inefficient Tools for Integrated Digital Twins Development. Security Gaps in Communication Channels. Weak Authentication in IoT Devices. Reconnaissance Attacks.
[49]	2019	IT	DLT	Data integrity, confidentiality, and security issues.
[50]	2021	IT	Digital shadows	Latency and real-time issues in digital shadows.
[51]	2022	IT	Data-driven modeling, ZTA	Risk assessment, and the seamless integration of innovative security solutions into existing large-scale systems.

Suhail et al. [53] Developed a platform called SecurINg CybEr-Physical sysTEm through dIgital twin-driven decep-tION (INCEPTION), for deceiving the attacker away from the real system. The idea of this paper is to mislead the attacker by using DT as a honeypot and drive them to a fake system. To lower the expense of the system and enhance the security, they use the fidelity concept. The fidelity of the DT system means how accurately the virtual twin mirrors the physical one. The fidelity of a system can be high, medium, or low. For example, high fidelity means highly realistic to the physical system, which will raise the cost of the system and reveal more information about the real system making it less

secure. In contrast, low fidelity means simpler construction and less reality, which reduces the expense and reveals less information, see Figure 4. The key point of the paper is using twisted fidelity that varies the level of the fidelity according to attacker behavior. After convincing the attackers to enter the fake environment and monitoring and analyzing their behavior, the authors used AI and ML to set the right level of fidelity. They applied their theory to a water system after explaining the idea of twisted fidelity and its consequences that are related to security, such as information leakage.

Prathiba et al. [54] Developed a framework called secure Federated Learning (FL) that enhances the

authenticity, integrity, and reliability of the Industrial Internet of Things (IIoT) environment. In order to prevent Sybil attacks, data poisoning attacks, and model poisoning attacks, the authors used DTs, blockchain technology with non-fungible token algorithms. The IIoT is a distributed environment that shares data through multiple nodes, and to protect these data the authors used the DT as a reference point to prevent manipulation of the data during the update process of the system. The precision of the framework showed 97% accuracy compared to the normal federated learning and 0.07 loss rate. They used MNIST dataset and custom server, the dataset is divided into 30 subsets equally.

Liu et al. [55] Developed a fault diagnosis-based DT case study for a wind turbine drive train using the SVM classification technique. The authors utilized the Unity 3D platform and enhanced the system with a particle swarm optimization least squares algorithm, achieving a 99.1% accuracy, which was higher than the result from using SVM alone. They outlined the classification steps, including feature extraction and system optimization, and discussed the pros and cons of the algorithm used.

Wang et al. [56] Developed a framework to detect insider threats caused by employees, including fraud, data theft, and system sabotage. Digital twin technology was used to mimic user behavior and predict future actions in a short time, which helped the authors monitor and analyze user behavior. After that, deep learning was used to create the model, and the dataset was trained using the Azure and Python platforms. The used dataset is CERT, which contains five files related to the company and its employees. The model showed higher performance compared to the existing work.

Shaikh et al. [57] Established a case study that assesses the security in digital twin systems. They divided the digital twin's layers into four layers: physical layer, communication layer, virtual layer, and application layer. After that, they defined each layer and its security risks and countermeasures as shown in Figure 5. Then they divided the attackers into two types, skillful attackers and unskilled attackers. For monitoring the unskilled attackers' behavior, they used the discrete-time Markov chain process, and for the skillful attackers, they used the Markov decision process. In the study, the first line of defense is encryption and authentication, and the second line is anti-malware and intrusion detection. They demonstrated their theory in a healthcare-based digital twin system and compared their work with other existing work. They used the probability of attack success and the cost of attack as performance measures. They concluded that when there is a high defense level, the probability of attack decreases and the cost of attack increases.

Gopal et al. [58] highlighted fundamental challenges in DT systems, emphasizing the necessity for cost-effective defect diagnosis, real-time feedback, and precise fault detection. The paper suggested employing a hybrid approach integrating the IoT with DT technology to tackle these challenges. Additionally, the authors examined the application of ML

techniques and LSTM arrays, and for integration of data, Zigbee network technology facilitates data connectivity among various sensors, operating as a foundational layer for information or enhancing existing data collection processes.

According to Potgantwar et al. [59] ML improved the security of DTs for Industry 4.0 applications. Specifically, different deep learning methods, such as artificial neural networks, SVM, logistic regression, and k-means, can help automate the model and decide the most efficient algorithm for a particular endeavor. Companies achieved secure business objectives by deploying more real-time and intelligent cyber-threat prevention, detection, and data analysis capabilities. The authors used a linear regression algorithm in their model. They utilized DT to solve a challenge and safeguard against substantial threats, including DDoS, within an environment.

Gupta et al. [60] explored the design and implementation of a hierarchical Federated Learning (FL)-based Anomaly Detection (AD) model tailored for smart health ecosystems. Their innovative model utilized the Federated Time Distributed (FEDTIMEDIS) Long Short-Term Memory (LSTM) approach to develop the AD model. To illustrate its applicability, the authors conducted a case study on Remote Patient Monitoring (RPM) for anomaly identification. They also introduced an implementation framework that incorporates DT and Edge Cloudlet Computing (ECC).

Sousa et al. [61] introduced the ELEGANT project, offering a novel strategy for implementing security monitoring and management mechanisms tailored for critical infrastructures through DTs. The article looked into the foundational aspects and difficulties associated with security in critical infrastructures, the application of DTs for assessing security and implementing mitigation strategies, and the outcomes of a distributed testbed validation of the ELEGANT off-site approach. The utilization of ML algorithms, a regression-based classifier, and a security analysis pipeline were detailed. Additionally, the paper explored the creation of precise digital twins designed for security and monitoring applications.

Akbarian et al. [62] introduced an innovative intrusion detection technique based on digital twins, employing a mix of a Kalman filter, a particle swarm optimization algorithm, and a Support Vector Machine (SVM) algorithm. Their approach aims to promptly identify and classify cyber attacks on Industrial Control Systems (ICSs) and Cyber-Physical Systems (CPSs), facilitating effective mitigation. The work addressed issues concerning the inherent vulnerability and insufficient security measures for CPS and ICS, which could potentially expose critical infrastructure to cyber threats. To tackle these challenges, the authors recommended a comprehensive strategy, including intrusion detection, testing and simulation, misconfiguration detection, and penetration testing.

Liu et al. [63] proposed a DT framework for an indoor safety management system. The framework employed

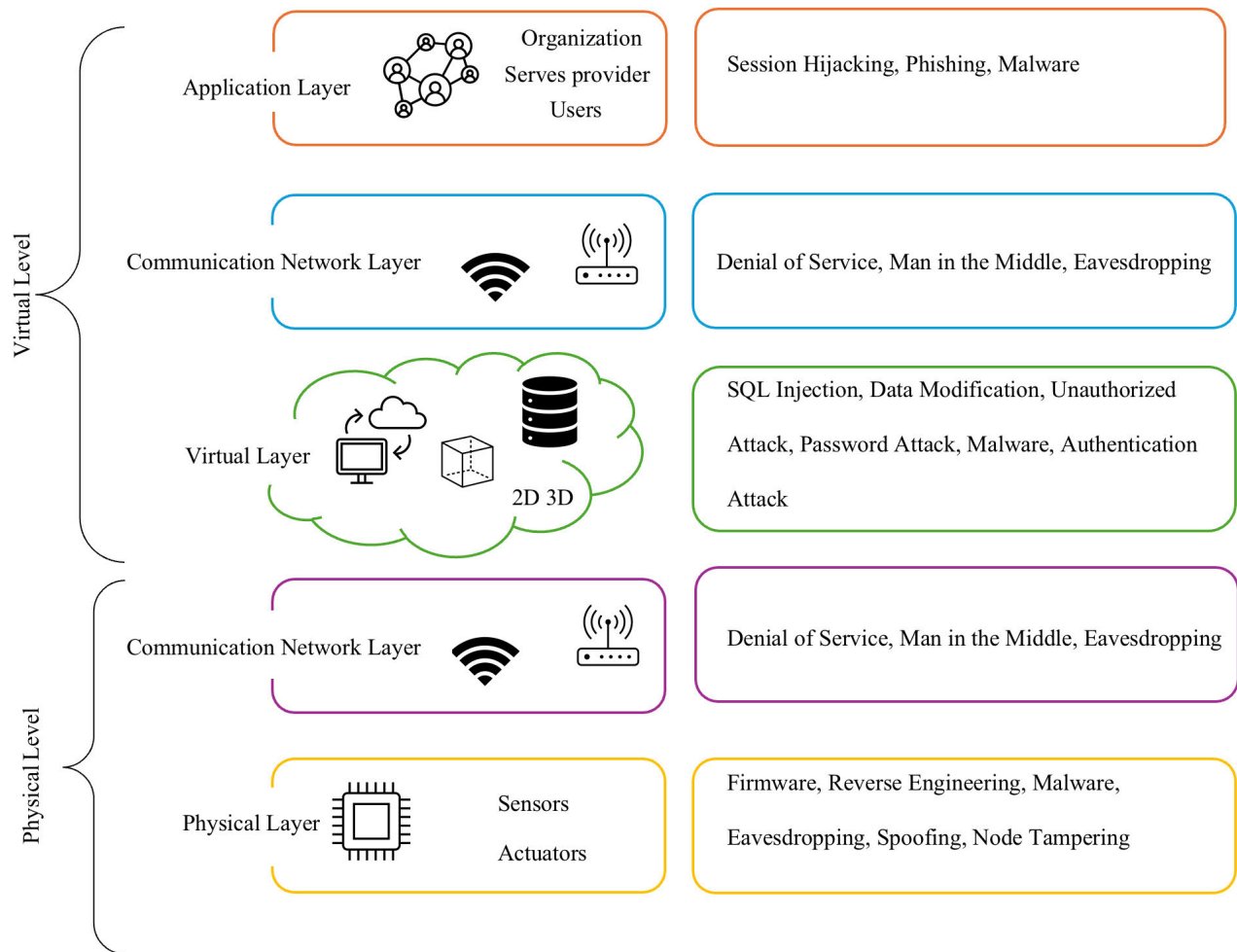


FIGURE 5. Digital twin layers and its security risks [57].

sensors and information analysis to fully analyze safety data and deliver real-time warnings and recommendations for improving safety. The paper discussed the importance of building an indoor safety system and how the proposed framework integrates building information modeling and IoT. The framework can be used to various types of facilities, such as smart homes or commercial buildings. The proposed framework is based on SVM for data analysis. The trained SVM model automatically obtained the types and levels of risk by processing the data in the DT model. The paper also discussed the data collection and processing techniques used in the study.

Kaur et al. [64] discussed several challenges in implementing DT technology, including cybersecurity, big data, data fusion, and standardization. They also highlighted the need for practical control systems to guarantee the security of the manufacturing system. The paper highlighted the importance of using ML in DT technology. It mentioned that ML is more helpful in analyzing real-time and offline data and extracting

valuable information from large amounts of dynamic and noisy data.

Kapteyn et al. [65] introduced an adaptable and effective method for constructing predictive digital twins, employing component-based reduced-order models and ML. The application of this approach was exemplified in a case study involving a self-aware drone, showcasing the development and continuous updating of a structural digital twin using sensor-derived data. The article also outlined the procedures for establishing the model library and employing optimal trees to train the data-driven classifier. In conclusion, the paper summarized the achieved results and outlines potential directions for future research.

V. ANALYSIS OF LITERATURE REVIEW

As per Figure 6, which illustrates the types of attacks identified in the collected papers, unauthorized access is the most common attack, followed by DoS attacks, data breaches, and malware attacks. Meanwhile, Figure 7 highlights the

TABLE 4. Summary of ML applications in DTs.

Ref.	Year	Sector/Case Study	Tool/Approach	Classification Result and Security Issues
[53]	2024	Water system	DTs and honeypot	Not specified. Security issues related to detecting different types of attacks.
[54]	2024	IIoT	Framework for securing FL-based DT using blockchain	Accuracy of 97%, Loss of 0.07. Security issues include sybil attacks, data poisoning, model poisoning.
[55]	2024	Wind turbine	Enhanced SVM for classification	Accuracy of 99.1%. Security issues include faults in wind turbines and difficulty in fixing due to remote locations.
[56]	2023	Organizations	DT, deep learning, NLP	Metrics include Accuracy, Precision, Recall, F1-score, AUC. Insider threats: fraud, data theft, system sabotage.
[57]	2023	Healthcare	Probabilistic model checking, Markov decision process	Attack success probability and cost. Security issues span multiple layers: Physical (eavesdropping, spoofing), Network (DoS, MITM), Virtual (SQL Injection, unauthorized access), Application (session hijacking, phishing).
[58]	2023	IT	LSTM and Zig-bee	Not specified. Security issues involve defect diagnosis, real-time feedback, fault detection.
[59]	2022	IT	ML, linear regression	Not specified. Security issues include DDoS threats.
[60]	2021	Healthcare	Hierarchical Federated Learning-based anomaly detection	Not specified. Security concerns involve patient data privacy, data poisoning, model drift.
[61]	2021	Critical infrastructure	ML algorithms, regression classifier	Normal condition: Precision 100%, Recall 96%; Attack condition: Precision 98%, Recall 100%. Security challenges in critical infrastructures.
[62]	2020	ICS	Matlab Simulink, Kalman filter, SVM	Accuracy: Ramp attack 91%, Scaling attack 98%, Normal conditions 99%. Security issues related to ICS connectivity.
[63]	2020	Smart homes	SVM	Accuracy of 99.25%, Precision of 97.07%, Recall of 99.38%. Security issues involve indoor safety management.
[64]	2020	IT	ML and AI	Not specified. Security concerns include cybersecurity, big data, data fusion, standardization.
[65]	2020	UAV	Component-based reduced-order models, ML using decision trees	Not specified. Security concerns involve predictive DT for UAV damage detection and rapid updates.

different sectors and the number of papers associated with each.

Figure 8 shows the most common techniques for detecting different types of attacks in DTs, where the techniques are divided into two types: ML techniques and non-ML techniques. See Figure 9, which represents the percentage of papers that use these two types. The non-ML techniques include statistical methods, rule-based systems, simulation techniques, optimization algorithms, expert systems, traditional data analytics, and mathematical modeling. On the other hand, the ML techniques are SVM, random forest, neural networks, and K-Nearest Neighbors.

A. EFFICIENCY OF CYBERSECURITY METHODS IN DIGITAL TWINS

This reviewed study highlights methodologies for protecting DTs, with different levels of efficiency in terms of threat detection, computational overhead, and adaptability. ML-based IDS, viz. SVM and DNN demonstrate high accuracy in detecting anomalies. However, these require significant computational resources, limiting real-time applications in resource-constrained environments. Encryption techniques, e.g., AES, TLS/SSL, ensure data confidentiality. However, these incur latency, which can be critical in real-time DT applications. Authentication protocols, including MFA and blockchain-based identity verification, enhance security but can complicate system integration. Furthermore, adaptive security frameworks, such as AI-driven risk assessment

models, offer dynamic threat mitigation. However, it requires continuous updates to remain effective. These findings depict that various cybersecurity techniques provide high resilience of DT systems. However, a balance between security, computational speed and complexity to deploy these techniques is to be maintained.

VI. COMPARISON WITH OTHER REVIEW PAPERS

This section shows how our paper differed from the existing review papers. The review paper by Kritzinger et al. [66] highlighted case studies of the different types of communication of DTs. While Enders et al. and Semeraro et al. [4], [5] included multiple definitions of DT across the years. In Jones et al. paper’s [6], focused mainly on characterizing DTs into 13 characteristics. Singh et al. [7] explained deeply the application of DT in different sectors. Somers et al. [67] reported carefully each step used in methodology used in the selection of their papers. On the other hand, Liu et al. [8] focused on the applications that discussed the lifecycle of DT such as design, manufacturing and services phases, while Nguyen and Adhikari [21] collected papers that combine DT with Building information modeling systems, for more details see Table 5

Our review’s main aim is to explore DT technology, security, and privacy, with a primary focus on analyzing the cybersecurity challenges in DT. We defined DT in detail, providing the reader the ability to understand this technology, how it works, and its various types. In our

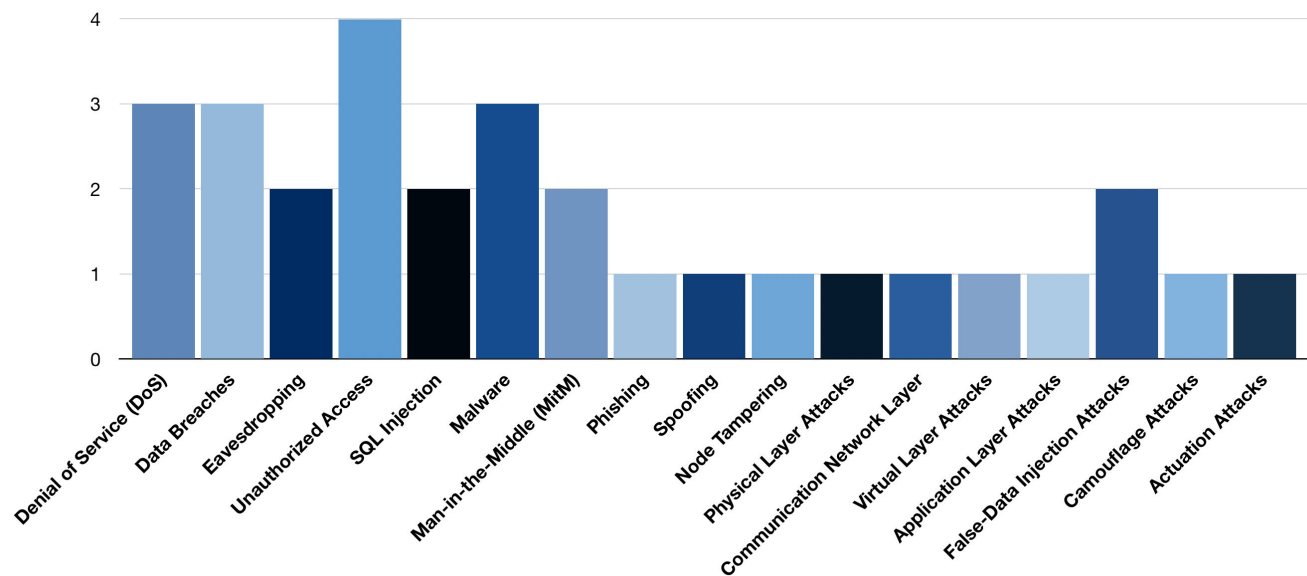


FIGURE 6. Type of attacks.

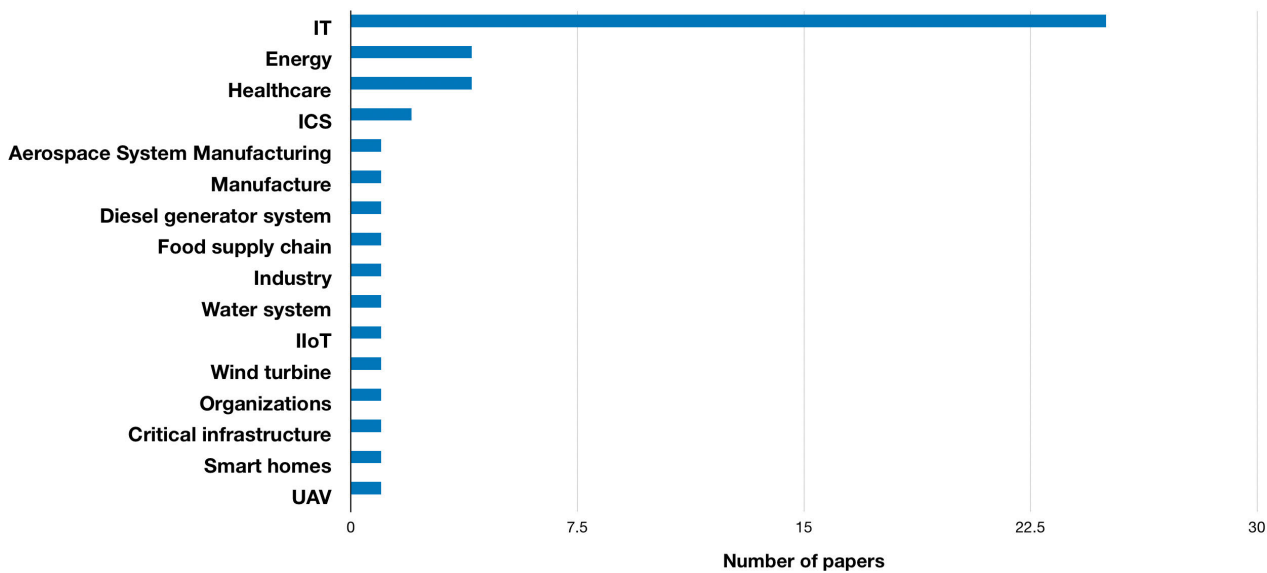


FIGURE 7. Different sectors utilizes DTs.

paper, we investigated different security issues with a detailed explanation. Furthermore, we categorise DT technology into different risk level. Moreover, the study aimed to highlight different case studies in different areas, paving the way for future research and practical applications in this domain. We presented an analysis of related studies by including case studies that involved issues like unauthorized access, data breaches, and DoS attacks. In addition, we included other case studies that used MLwith DT to strengthen security. While other review papers did not explore similar issues regarding cybersecurity, and they did not clarify limitations

of implementing DT technology. Furthermore, our review presented an analysis of related studies, identified existing limitations, and shed light on prospective future directions in DT.

VII. FUTURE WORK AND OPEN CHALLENGES

In this section, we explore diverse avenues for the future of DTs from various perspectives. The advancement of DT technology for secure and efficient operations brings various challenges and areas for future exploration. Integrating emerging technologies, such as blockchain and advanced

TABLE 5. Comparison with other review papers.

Reference	[66]	[4]	[6]	[5]	[8]	[7]	[67]	[21]	Our review
Year	2018	2019	2020	2021	2021	2022	2023	2023	2024
Number of Papers Included in The Review	-	78	92	115	-	-	26	30	37
Case Studies	✓				✓	✓	✓	✓	✓
Limitations of Other Studies								✓	✓
Studies Include ML						✓			✓
History			✓		✓	✓		✓	✓
Risk Assessment									✓
DT Sectors and Areas		✓	✓	✓		✓	✓	✓	✓
Systematic Literature Methodology	✓		✓	✓	✓	✓	✓	✓	✓

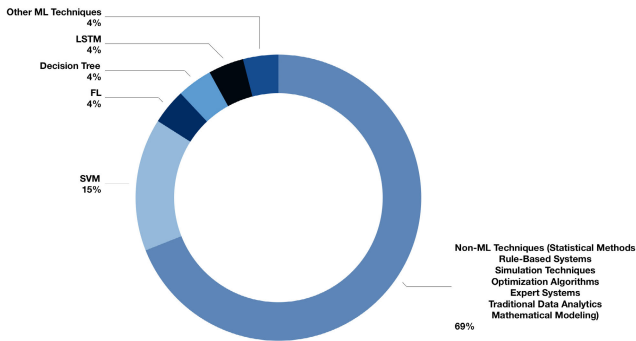


FIGURE 8. Analysis of research techniques: ML vs. Non-ML approaches.

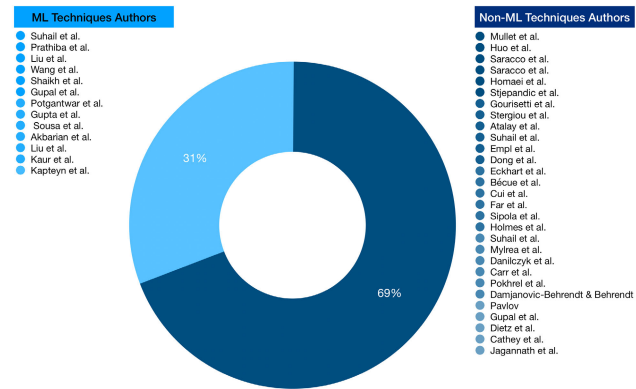


FIGURE 9. Proportion of research papers utilizing ML vs. Non-ML techniques.

encryption methods, requires further investigation to enhance the overall security posture of DT systems. Scalability remains a critical consideration, encouraging researchers to develop solutions adaptable to the complexities of large-scale implementations. Real-time adaptation and resilience mechanisms demand attention to ensure DT security measures can dynamically respond to evolving threats. Standardization and establishing best practices are essential to creating a unified framework for strong security across diverse DT applications.

Time and cost are still factors, as implementing DT systems, including the physical components, software, and specifically talented staff, are expensive to develop and, in many instances, beyond the reach of small businesses. However, the technical challenges of building an exact digital twin model and integrating it with the physical environment lead to other organizational factors that may delay adoption rates across different industries.

User authentication and access control mechanisms need refinement, exploring multi-factor authentication and biometric measures for boosted security. Addressing privacy concerns and navigating the legal implications of DT data becomes imperative as industries deal with stringent data privacy regulations [68]. Cross-disciplinary collaboration among cybersecurity experts, domain specialists, and policy-makers is encouraged to foster comprehensive and effective security solutions [69]. Integration issues also come into the picture, because usually DTs have to run in parallel with existing systems as well as state-of-the-art technologies like IoT and AI. This absence of compatibility raises the system risks and challenges the processes to implement the use in energy, healthcare, and manufacturing industries. Domain-related issues likewise define the practical usage of DTs: For example, when applied in healthcare, DTs are applied in predictive diagnostics and patient monitoring, but data privacy becomes an issue when working under HIPAA rules. Likewise, DTs play an important role from a technical perspective when it comes to predictive maintenance for equipment and product manufacturing processes; however, the challenge lies in implementing these DT environments with legacy equipment and/or complex working patterns. In the energy sector, DTs can be used to keep track of and adjust grid stability, although there are always risks of cyberattacks against critical facilities. For smart cities, DT is significant for resource management and urban planning, but the incorporation of real-time data from numerous sources concerning privacy is complex. Last in construction, DTs apply to the project management and the safety monitoring of the project but as with all DTs they are expensive and require accurate tools for making the model. In addition, the

development of ZTA as a practical use case warrants further exploration for rapid prototyping, testing, and integration of innovative security solutions into existing large-scale systems. This comprehensive approach aims to inspire researchers and practitioners to contribute to the evolving digital twin security landscape, ensuring the robustness and resilience of these transformative technologies.

From the standpoint of *Data Security*, safeguarding against attacks on generated data or modeling methods is crucial, including mitigating evasion and poisoning attacks that manipulate DT models for false positives. The absence of comprehensive datasets for digital twins emerges as a significant impediment, demanding efforts to overcome this limitation in research and development [70].

Concerning *Control System Security*, the reliability of DT software for autonomous control systems is essential, considering the escalating use of DT software in directly controlling physical systems. Security vulnerabilities in such software can have profound consequences, necessitating comprehensive strategies for robust protection [71].

For *IoT Security*, seamless integration of IoT with cybersecurity in digital twin and future standards is pivotal. Merging established standards with cybersecurity concepts is essential to enhance real-time data integration, establish robust cybersecurity frameworks, develop smart application architectures, and institute dedicated cybersecurity layers to safeguard assets in the evolving built environment. Automation in identifying, quantifying, and reevaluating cyber risks based on digital twins is a critical research focus [72]. Additionally, exploring the efficacy of honeypots and behavioral alterations in digital twins for data protection while maintaining realism requires further investigation.

Physical System Security involves monitoring physical aspects of CPSs through DTs for intrusion detection, a research direction that remains largely unexplored. The exploration of intrusion prevention mechanisms within digital twins, especially in real-time scenarios, is deemed essential.

Control System and IoT Security present challenges in understanding how to introduce latency effectively to maintain security while meeting real-time requirements. Moreover, addressing introduced latency in incident response training, particularly in real-time scenarios, is vital for creating effective and realistic training environments.

Some specific examples of threats DTs face include model poisoning attacks, where attackers inject malicious data into DT training data, leading to incorrect predictions. Model evasion attacks involve crafting input data to fool DT models, while data injection attacks inject malicious data into the data stream, causing incorrect predictions or loss of control [73]. Additionally, software vulnerabilities in DT software can be exploited to gain control or disrupt operations [74]. The overarching goal is to shape the future of DTs by addressing these challenges comprehensively and guaranteeing the security, reliability, and resilience of these transformative technologies.

VIII. CONCLUSION

This systematic literature review has shown the vast landscape of digital twins, drawing insights from 74 existing research publications that rigorously address challenges and techniques within this transformative technology. The analyzed publications span various sectors, including energy, healthcare, IT, ICS, aerospace, and critical infrastructure, employing multiple techniques such as statistical methods, rule-based systems, simulation techniques, optimization algorithms, expert systems, traditional data analytics, mathematical modeling, and ML algorithms. The diversity in tools reflects the multidisciplinary nature of DT research, with each tool addressing specific challenges and contributing to the advancement of the field. The security concerns associated with DTs are multidimensional, encompassing issues like unauthorized access, data breaches, DoS, cyberattacks, and vulnerabilities in communication networks. The review underscores the significance of addressing security gaps ensuring confidentiality, integrity, and availability of digital replicas. Additionally, challenges such as the lack of transparency, interoperability concerns, and the potential misuse of artificial intelligence within DT ecosystems are highlighted. Furthermore, the literature points to specific security risks in sectors like healthcare, aerospace, and smart homes, emphasizing the significance of tailored cybersecurity measures for diverse applications of DTs. In conclusion, this comprehensive review shows the state-of-the-art in digital twins and identifies security challenges and mitigation strategies. The insights from these publications provide a valuable foundation for researchers, practitioners, and decision-makers navigating the evolving landscape of DTs and their associated security concerns.

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